Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education, Vol 12(4), pp. 618-627; www.cwsei.ubc.ca

Date: _____ Class: _____ Instructor: _____ No. students _____ Observer Name: _____

Classroom arranged how? (include any special notes about environment) _____

1. **L**-Listening; **Ind**-Individual work; FoI - Follow-along worksheet; **CG**-Clicker Q discussn; **WG**-Worksheet group work; **C**-Writing code on computer; **OG**-Other group work; **AnQ**-Answer Q; **SQ**-Student Q; **WC**-Whole class discuss.; **Prd**-Predicting; **SP**-Student present.; **TQ**-Test/quiz; **W**-Waiting; **O**-Other

2. **Lec**-Lecturing; **RtW**-Writing; **FUp**-Follow-up; **PQ**-Pose Q; **CQ**-Clicker Q; **AnQ**-Answer Q; **MG**-Moving/Guiding; **1o1**-One-on-one; **D/V**-Demo+; **Adm**-Admin; **W**-Waiting; **O**-Other

3. **L** - low engagement (observer confident majority of students are not engaged); **M** - medium engagement (observer is unsure); **H** - high engagement (observer is confident majority of students are engaged); **InterI** - number of unique students instructor interacted with; **InterT** - number of unique students TA(s) interacted with

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

	1. Students doing															**2. instructor doing**												**3. Engagement**					**Comments:** explain difficult coding choices
min	L	Ind	FoI	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
0 - 2																																	
2																																	
4																																	
6																																	
8 - 10																																	
	L	Ind	FoI	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
10 - 12																																	
12																																	
14																																	
16																																	
18 - 20																																	
	L	Ind	FoI	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
20 - 22																																	
22																																	
24																																	
26																																	
28 - 30																																	

	1. Students doing															2. instructor doing												3. Engagement					Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Interl	InterT	
30 - 32																																	
32																																	
34																																	
36																																	
38 - 40																																	
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Interl	InterT	
40 - 42																																	
42																																	
44																																	
46																																	
48 - 50																																	
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Interl	InterT	
50-52																																	
52																																	
54																																	
56																																	
58-60																																	

	1. Students doing															2. instructor doing												3. Engagement					Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Interl	InterT	
60-62																																	
62																																	
64																																	
66																																	
68-70																																	
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Interl	InterT	
70-72																																	
72																																	
74																																	
76																																	
78-80																																	
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Interl	InterT	
80-82																																	
82																																	
84																																	
86																																	
88-90																																	

	1. Students doing															2. instructor doing												3. Engagement					Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Interl	InterT	
90-92																																	
92																																	
94																																	
96																																	
98-100																																	
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Interl	InterT	
100-102																																	
102																																	
104																																	
106																																	
108-110																																	
Summary																																	